ALEXIS CORPORAL

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PROFESSIONAL SUMMARY

Full-Stack Developer and Cum Laude graduate with a Bachelor of Science in Information Technology, experienced in developing secure, scalable software for educational institutions, manufacturing, and business operations. Proficient in designing backend services, relational databases, and modern web applications using Go, TypeScript, Python, and PostgreSQL. Passionate about building reliable software that automates workflows and solves real-world problems.

TOOLS & TECHNOLOGIES

Languages: TypeScript, Go, Python, Java, SQL, Dart, JavaScript

Frameworks: Next.js, Fastify, FastAPI, Spring Boot, React, Flutter, Prisma

Infrastructure & Database: Docker Compose, Nginx, PostgreSQL, MySQL, Supabase, WebSockets, Tesseract OCR

Testing/Tools: Locust, Postman, JMeter, Neovim, Git, VS Code

WORK EXPERIENCE

Freelance Full-Stack Developer May 2023 - Present

- Architected relational database schemas for inventory tracking, appointment scheduling, and document management systems used in educational, manufacturing, and business environments.
- Automated manual inventory operations by building a QR code generation pipeline and transactional backend services for a school inventory management system.
- Developed a multi-role veterinary management system as a Progressive Web App (PWA) with role-based access control and service worker–based caching to provide offline access to previously loaded images and application resources.

Bakawan Data Analytics, Inc. - *Web Developer Intern (Awarded 3rd Best System out of 9 teams)* Feb 2026 - May 2026

- Collaborated within a dedicated engineering team to translate core system blueprints into scalable web applications.
- Developed backend authorization logic including role validation, secure state transitions, and audit logging.
- Verified API performance and stability using automated JMeter and Postman test suites.

KEY PROJECTS

Digital Information Board Management System (DIBMS) Jul 2026 - Present

Next.js, TypeScript, Tailwind CSS, Prisma, Supabase (PostgreSQL + Storage)

- Implemented a role-based access control (RBAC) system supporting four organizational roles (System Admin, Coordinator, Elder, Ministerial Servant) to enforce secure access across the platform.
- Engineered an asset automation and parsing pipeline using pdfjs-dist to isolate and rasterize multi-page vector PDFs client-side, offloading processing overhead and delivering immediate interface rendering.
- Designed a normalized schema with 9 relational tables using Prisma-managed foreign keys to ensure transactional consistency across congregation schedules, audit logs, and section assignments.
- Built an event-driven public display interface that auto-detects active congregations based on local timezone schedules, featuring a manual override with automated timeout returns for large TV display deployments.

DLSP Registrar Docs Track Feb 2026 - Jun 2026

Go, Next.js, React, TypeScript, PostgreSQL, Gorilla Mux & WebSocket

- Developed a full-stack document tracking system for the College Registrar's Office using a Go API secured with middleware-based rate limiting.
- Architected and implemented a strict 7-stage state machine request workflow (from initial submission through payment tracking to ready for pickup and complete) to govern institutional records securely.
- Built a duplex communication subsystem using Gorilla WebSocket to stream instantaneous status updates and live notifications to connected clients without browser polling.
- Structured a robust JWT authentication layer featuring access/refresh token rotation and an administrative audit trail tracking user actions along with IP and user-agent metadata.

BakerPass Aug 2025 - Jan 2026

Fastify, Next.js, Flutter, MySQL, Nginx, Docker Compose

- Orchestrated a containerized multi-service environment using Docker Compose to isolate and run the frontend, a Fastify API server, and a persistent MySQL database.
- Architected a multi-tenant visitor management lifecycle incorporating a 6-role permission system (Visitor, Employee, Security, HR, Nurse, Super Admin) to govern workflows for protocol-enforced factory environments.
- Configured an Nginx reverse proxy layer handling centralized request routing, automated WebSocket protocol upgrades, and connection rate limiting on authentication endpoints to secure backend nodes against brute-force traffic.
- Built a live synchronization layer using Fastify's WebSocket subsystem to broadcast immediate appointment updates and health declaration statuses simultaneously across a web dashboard and a Flutter mobile companion app.

Engineering Drawing OCR Extractor Feb 2026 - Jun 2026

Python, FastAPI, Tesseract OCR, pypdf, PyMuPDF, Locust

- Engineered a hybrid text extraction engine combining lightweight vector PDF parsing (executing in under 0.5s) with a Tesseract OCR fallback layer to seamlessly ingest dense engineering documentation.
- Optimized layout precision and reduced graphic noise by implementing horizontal band scanning constraints (restricting processing to top-bottom / 20% title blocks) and runtime multi-region coordinate cropping.
- Improved throughput for CPU-bound image rasterization by using ProcessPoolExecutor to bypass Python's Global Interpreter Lock (GIL).
- Structured an automated benchmarking harness with Locust to profile backend capabilities under sustained load, mapping hardware performance ceilings during high-resolution 300 DPI processing.

EDUCATION

Dalubhasaan ng Lungsod ng San Pablo (DLSP) 2022 - 2026

Bachelor of Science in Information Technology (Software Development)
Graduated Cum Laude

TRAINING & CERTIFICATIONS

- **Best in OJT & Best Capstone Project Awards** - Recognized for outstanding performance in Software Development (2026)
- **AWS Certified Cloud Quest: Cloud Practitioner** - AWS Training & Certification, March 2026